P1 电源树

File: demo_mainboard_p1.kicad_sch

P2 SoC 主控

File: demo_mainboard_p2.kicad_sch

P3 启动存储

File: demo_mainboard_p3.kicad_sch

P4 显示双出口

File: demo_mainboard_p4.kicad_sch

P5 MIPI-CSI 传感器

File: demo_mainboard_p5.kicad_sch

P6 高速差分与连接器

File: demo_mainboard_p6.kicad_sch

P7 板级接口与测试点

File: demo_mainboard_p7.kicad_sch

P8 时钟·复位·调试

File: demo_mainboard_p8.kicad_sch

P9 跨铰链界面

File: demo_mainboard_p9.kicad_sch

P10 充电板(镜框)

File: demo_mainboard_p10.kicad_sch

each sheet = one CSV page P1..P10
10 functional sheets · pins from HW/netlist_v1_pins.csv
self-drawn, not ODM

Sheet: /
File: demo_mainboard.kicad_sch

Title: DEMO MAINBOARD — SCHEMATIC INDEX / 原理图目录

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 1/11

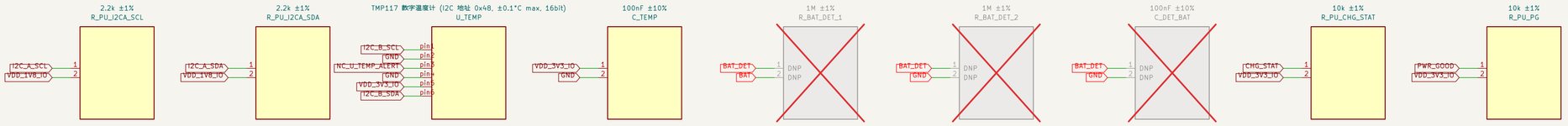
pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 2 · 65 devices · paper A2
self-drawn, not ODM

Sheet: /P1 电源树/
File: demo_mainboard_p1.kicad_sch

Title: P1 电源树 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11
KiCad 8.0.4, 10.0.8	Id: 2/11

Rev: draft-3

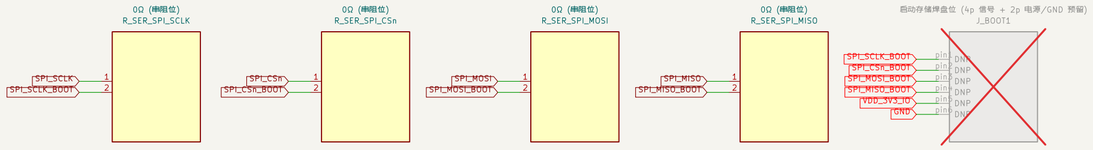


pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 3 · 9 devices · paper A2
self-drawn, not ODM

Sheet: /P2 SoC 主控/
File: demo_mainboard_p2.kicad_sch

Title: P2 SoC 主控 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 3/11

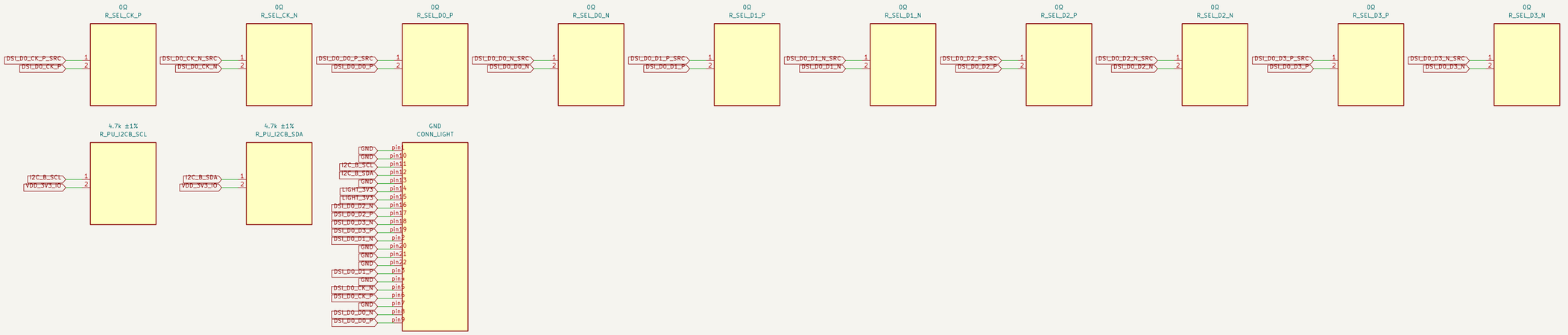


pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 4 · 5 devices · paper A2
self-drawn, not ODM

Sheet: /P3 启动存储/
File: demo_mainboard_p3.kicad_sch

Title: P3 启动存储 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 4/11

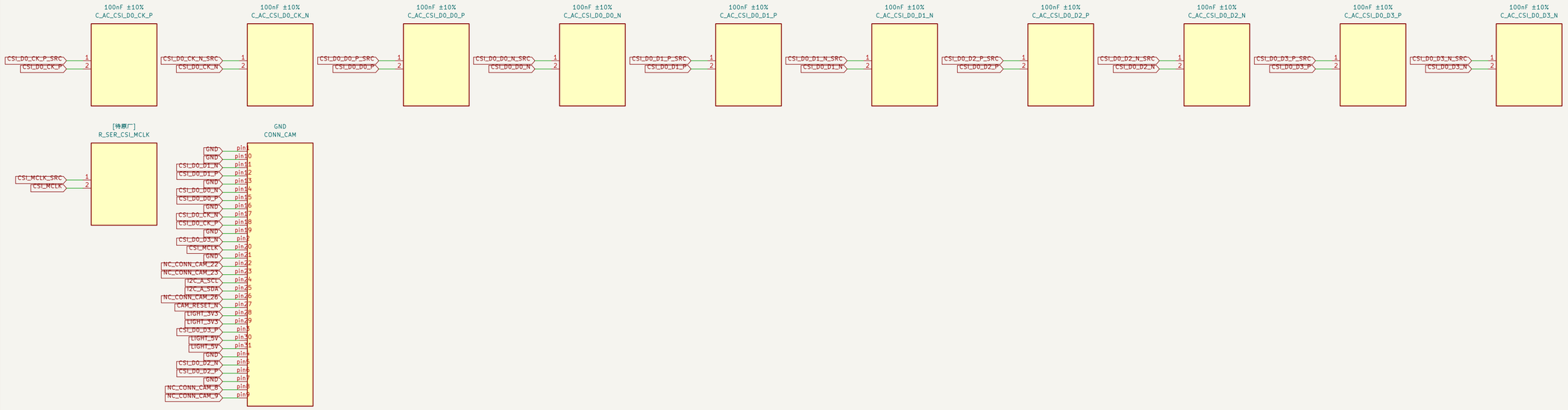


pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 5 · 13 devices · paper A2
self-drawn, not ODM

Sheet: /P4 显示双出口/
File: demo_mainboard_p4.kicad_sch

Title: P4 显示双出口 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 5/11

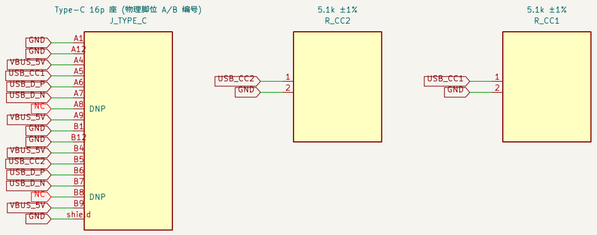


pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 6 · 12 devices · paper A2
self-drawn, not ODM

Sheet: /P5 MIPI-CSI 传感器/
File: demo_mainboard_p5.kicad_sch

Title: P5 MIPI-CSI 传感器 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 6/11

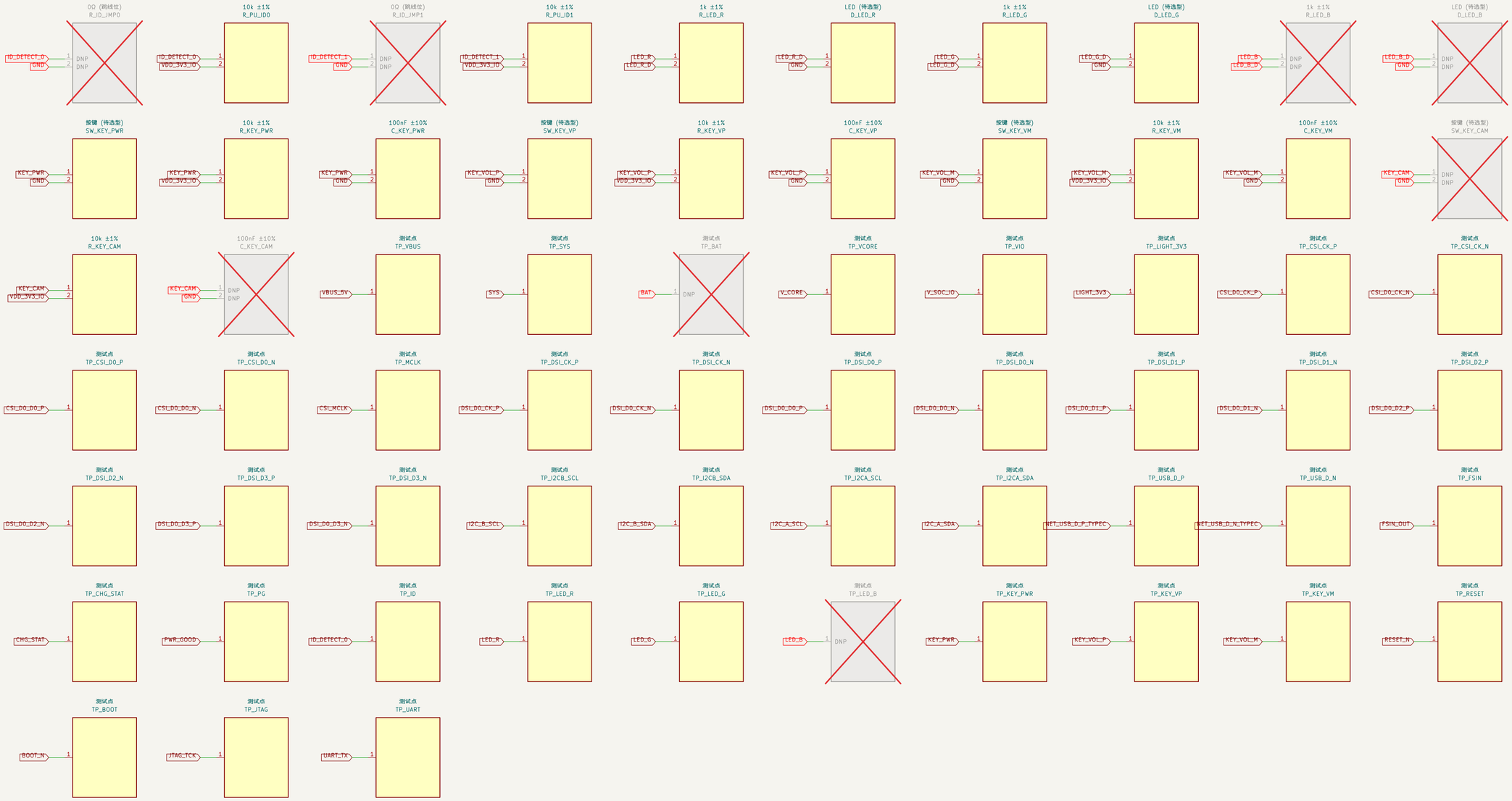


pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 7 · 3 devices · paper A2
self-drawn, not ODM

Sheet: /P6 高速差分与连接器/
File: demo_mainboard_p6.kicad_sch

Title: P6 高速差分与连接器 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 7/11



pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 8 · 63 devices · paper A2
self-drawn, not ODM

Sheet: /P7 板级接口与测试点/
File: demo_mainboard_p7.kicad_sch

Title: P7 板级接口与测试点 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 8/11

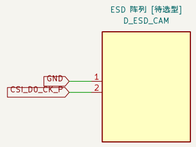
pins from HW/netlist_v1_pins.csv (58-pin truth) · net = global label
sheet 9 · 28 devices · paper A2
self-drawn, not ODM

Sheet: /PB 时钟·复位·调试/
File: demo_mainboard_pb.kicad_sch

Title: PB 时钟·复位·调试 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11
KiCad E.D.A. 10.0.8	Rev: draft-3
	Id: 9/11

R_PG_PULL
10k ±1%
PG_AND
VDD_IVB_IO
R_EN_VCORE
10k ±1%
EN_VCORE
VDD_IVB_IO
Q_AND
[待选型] N-MOS (硬件 AND)
RESET_N
PG_AND
R_PU_RESET_N
10k ±1%
RESET_N
VDD_IVB_IO
R_SER_RESET_N
1k ±1%
RESET_N
RESET_N_SOC
C_RST
100nF ±10%
RESET_N_SOC
GND
SW_RST
复位按钮 (待选型)
RESET_N
GND
R_PU_BOOT_N
10k ±1%
BOOT_N
VDD_IVB_IO
SW_BOOT
BOOT 按键 (待选型)
BOOT_N
GND
R_JTAG_TCK
1000 ±1%
JTAG_TCK_SOC
JTAG_TCK
R_JTAG_TMS
1000 ±1%
JTAG_TMS_SOC
JTAG_TMS
R_JTAG_TDI
1000 ±1%
JTAG_TDI_SOC
JTAG_TDI
R_JTAG_TDO
1000 ±1%
JTAG_TDO_SOC
JTAG_TDO
J_DEBUG
1×4 1.27mm 排针 (JTAG)
JTAG_TCK
JTAG_TMS
JTAG_TDI
JTAG_TDO
J_UART
1×2 1.27mm 排针 (UART)
UART_TX
UART_RX
R_SER_FSIN
50Ω ±1%
FSIN_OUT
FSIN
TP_PG_VCORE
PG (开漏, 需上拉)
PG_AND
TP_PG_VIO
PG (开漏, 需上拉)
PG_AND
R_SUPV_DIV1
[待选型] 分压上臂 (0Ω 跳线位)
V_SOC_IO
SUPV_SENSE
R_SUPV_DIV2
[待选型] 分压下臂 (0Ω 跳线位)
SUPV_SENSE
GND
C_SUPV
100nF ±10%
VDD_IVB_IO
GND
U_DCDC_VCORE
EN 输入 (高 = 使能)
BOARD_5V
EN_VCORE
PG_AND
U_DCDC_VIO
EN 输入 (高 = 使能)
BOARD_5V
PWR_ON
PG_AND
Q_EVC
[待选型] N-MOS (反相)
PG_AND
EN_VCORE
U_SUPV
开漏释放
RESET_N
SUPV_SENSE
VDD_IVB_IO
C_XTAL_IN
12pF ±10%
XTAL_IN
GND
C_XTAL_OUT
12pF ±10%
XTAL_OUT
GND
X1
晶振焊盘位 (不装)
XTAL_IN
XTAL_OUT
XTAL



pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 10 · 1 devices · paper A2
self-drawn, not ODM

Sheet: /P9 跨较链界面/
File: demo_mainboard_p9.kicad_sch

Title: P9 跨较链界面 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 10/11

5.24k ±1% (RT1: REGN 到 TS)
R_TS_TOP
REGN
TS_BAT
1
2
DNP
DNP

1µF X7R 25V
C_VBUS_IC
VBUS_5V_CHG
PGND
1
2

10µF X7R 25V
C_PMID
PMID
PGND
1
2

2.2µH Isat≥1.5A DCR≤50mΩ
L_CHG
PMID
SW
1
2

47nF 25V
C_BST
BTST
SW
1
2

10k ±1%
R_PSEL
PSEL
PGND
1
2

10k ±1%
R_PU_SCL
I2C_CHG_SCL
REGN
1
2

10k ±1%
R_PU_SDA
I2C_CHG_SDA
REGN
1
2

10k ±1%
R_PU_INT
INT_CHG
REGN
1
2
DNP
DNP

10k ±1%
R_OTG
OTG_SET
PGND
1
2

10k ±1%
R_CE
CE_SET
PGND
1
2

10µF X7R 6.3V
C_BAT
BAT_CHG
PGND
1
2

20µF 6.3V
C_SYS
SYS_CHG
PGND
1
2
DNP
DNP

0Ω / 铜皮短接 (PGND 单点星接)
R_STAR_PGND
PGND
GND
1
2

5V 级单向 TVS [待选型]
D_TVS_VBUS
VBUS_5V
GND
1
2

测试点
TP_TS
TS_BAT
1

1k ±1% (按 LED Vf 定)
R_CHG_LED
VBUS_5V_CHG
LED_CHG_A
1
2

LED (待选型，充电板侧)
D_CHG_LED
LED_CHG_A
CHG_LED_K
1
2

[待选型] N-MOS
Q_CHG_LED
CHG_LED_K
GND
CHG_STAT
1
2
3

pins from HW/netlist_v1_pins.csv (584-pin truth) · net = global label
sheet 11 · 19 devices · paper A2
self-drawn, not ODM

Sheet: /P10 充电板(独板)/
File: demo_mainboard_p10.kicad_sch

Title: P10 充电板(独板) — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-3**
KiCad E.D.A. 10.0.8		Id: 11/11